

Configuration File (YAML)

engines · PE placement · memory technology · network · power

Workloads (Shared Memory / Message Passing)

μkernels · BabelStream · Rodinia · NPB · SPLASH-3 · PARSEC

Single simulation process — host & device accounting domains (QEMU user-mode + ZSim)

Host Simulation Engine

Core

Core

Core

Core

core_type: ooo / in-order / simple

Caches

L1+L2

Network

GARNET

Memory Controller

Event Queue

offload region (ROI)

host → device

device → host

same-thread
domain transition

transfers charged on
link + memory models
(PCIe · CXL · interposer)

Device Simulation Engine

PE

PE

PE

PE

pe_type: ooo / in-order / simple / alu / null

placement: subarray ▶ bank ▶ bank-group ▶ chip ▶ rank ▶ logic-die

Address Translation

TLB + page table

unified / discrete

In-memory Network

GARNET

(hierarchical H-tree for DRAM)

Modeling Infrastructure

Common Interface for External Models (YAML)

DRAM (Ramulator2)

DDR3/4/5 · LPDDR5 · GDDR6 · HBM2/3

SRAM (CACTI)

on-die / cache arrays

NVM (NVSim)

STT-MRAM · PCM · ReRAM

Compute (ZSim / QEMU)

ooo · in-order · simple · alu · null

Network (GARNET)

H-tree · mesh · torus · crossbar · fat-tree
accuracy: analytical / detailed

Power & Area (McPAT)

cores · caches · NoC · PCIe